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A Data-Driven Ticketing System for Optimized Anti-Rabies Vaccination Management

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This work is lovingly dedicated to:

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whose sacrifices, patience, and unconditional love gave us the strength to persevere through every challenge.

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whose wisdom and dedication lit the path of our learning and inspired us to aim higher.

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And to every patient who seeks care at the ABTC—

may this system serve as a small but meaningful contribution to your safety and well-being. —

Isidro, Daclis, and Tinaan

ABSTRACT

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N-VISION: A Data-Driven Ticketing System for Optimized Anti-Rabies Vaccination Management

Bachelor of Science in Information Systems, Mandaluyong College of Science and Technology, March 2026

Rabies remains a significant public health threat in the Philippines, with the Mandaluyong City Animal Bite Treatment Center (ABTC) serving as a frontline facility for post-exposure prophylaxis (PEP) and anti-rabies vaccination services. The current manual and fragmented operational system of the ABTC presents critical inefficiencies in patient registration, vaccination scheduling, inventory tracking, and reporting compliance with the Department of Health (DOH) and the Local Government Unit (LGU).

This study proposes the design, development, and evaluation of N-VISION: A Data-Driven Ticketing System for Optimized Anti-Rabies Vaccination Management. The system integrates six core modules: (1) patient registration, (2) ticket queue management, (3) vaccination scheduling, (4) vaccine inventory monitoring, (5) automated SMS reminders, and (6) a reporting dashboard. The system was developed using an Agile-Scrum methodology and evaluated against ISO/IEC 25010 software quality standards, covering functional suitability, performance efficiency, usability, reliability, and security.

Data collection involved requirements analysis with ABTC staff, pilot deployment, and structured evaluation through User Acceptance Testing (UAT) and ISO/IEC 25010 questionnaires administered to eight respondents, including nurses, doctors, administrative staff, and IT experts. Survey responses were analyzed using descriptive statistics, including frequency distribution, percentage, and weighted mean.

The system addresses identified gaps in the literature, particularly the lack of integrated ticketing and scheduling systems tailored for animal bite treatment centers in the Philippine local government context. The findings are expected to demonstrate that the proposed system can significantly improve operational efficiency, reduce patient waiting times, enhance vaccination compliance, and support accurate and timely reporting to health authorities

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LIST OF ABBREVIATIONS

ABTC Animal Bite Treatment Center

BSCS Bachelor of Science in Computer Science

BSIS Bachelor of Science in Information Systems

CHO City Health Office

DOH Department of Health

GIS Geographic Information Systems

ISO International Organization for Standardization IEC International

Electrotechnical Commission

IT Information Technology

LGU Local Government Unit

MCST Mandaluyong College of Science and Technology PEP Post-Exposure Prophylaxis

RA Republic Act

SMS Short Message Service

UAT User Acceptance Testing

WHO World Health Organization

CHAPTER 1

Introduction

Rabies is a fatal yet preventable viral disease that continues to pose a serious public health threat worldwide. Despite the availability of safe and effective vaccines, rabies causes an estimated 59,000 human deaths annually, with the highest burden concentrated in Asia and Africa (Hampson et al., 2022; Chen et al., 2025). The persistence of the disease is often linked not to medical limitations, but to systemic challenges in healthcare delivery, public awareness, and animal control programs.

In the Philippines, rabies remains classified as a public health problem. The Department of Health reports cases each year, emphasizing that deaths occur primarily due to delayed or incomplete post-exposure prophylaxis and lack of immediate access to Animal Bite Treatment Centers or ABTCs (Department of Health Philippines, 2022). These centers are critical frontline facilities responsible for assessing bite cases, administering anti-rabies vaccines, and ensuring that patients complete their required immunization schedules.

The effectiveness of ABTCs, however, is heavily dependent on the efficiency of their operational workflows. In Mandaluyong City, the local ABTC serves a large and growing population, resulting in high patient volumes that intensify the limitations of existing manual systems. Paper-based registration, manual scheduling of follow-up doses, fragmented inventory tracking, and labor-intensive report generation contribute to longer waiting times, data inaccuracies, and increased risk of missed vaccinations (Luna & Reyes, 2022; Mandaluyong City Health Office, 2023). These inefficiencies not only affect patient care but also hinder compliance with national reporting standards required by the Department of Health.

Advancements in digital health systems have demonstrated significant potential in addressing similar challenges in public health settings. Studies show that information systems can streamline vaccination management, reduce human error, and improve health service delivery through automation, real-time monitoring, and data integration (Nguyen et al., 2022). Applying these solutions to ABTCs offers an opportunity to modernize anti-rabies services and strengthen local implementation of the National Rabies Prevention and Control Program.

In response to these concerns, this study proposes the development of a data-driven ticketing

system for the Mandaluyong City ABTC. The system aims to integrate patient registration, ticket queue management, vaccination scheduling, inventory monitoring, automated SMS reminders, and a reporting dashboard into a unified platform. Through this intervention, the study seeks to improve operational efficiency, enhance patient compliance with vaccination schedules, and support accurate and timely reporting to health authorities.

This study is aligned with the United Nations Sustainable Development Goals, particularly SDG 3: Good Health and Well-Being and SDG 16: Peace, Justice, and Strong Institutions. The study holds significant relevance to SDG 3 because it supports the improvement of anti-rabies vaccination services by organizing patient registration, ticket queue management, vaccination scheduling, inventory monitoring, and SMS reminders. Through these features, the system can help reduce missed vaccination schedules, improve patient flow, and support timely completion of anti-rabies vaccination doses.

The study is also aligned with the implementation of Republic Act No. 11032, also known as the Ease of Doing Business and Efficient Government Service Delivery Act of 2018, which promotes the digitalization and streamlining of government services to improve efficiency, transparency, and public service delivery. Through a data-driven ticketing system with dashboards, reports, queue monitoring, and automated scheduling features, the system supports LGU healthcare digitalization initiatives and enhances the operational efficiency of Animal Bite Treatment Center services. Additionally, the system considers the principles of Republic Act No. 10173, or the Data Privacy Act of 2012, by promoting secure handling and management of patient information and vaccination records.

Background of the Study

Rabies continues to cause an estimated 59,000 deaths worldwide each year, with most cases occurring in Asia and Africa (Hampson et al., 2022; Chen et al., 2025). Although effective vaccines and post-exposure prophylaxis are available, the disease remains a major public health concern due to gaps in public awareness, animal control measures, and access to healthcare services.

In the Philippines, rabies remains a persistent health issue, with cases reported annually despite the availability of vaccines (Department of Health Philippines, 2022). Contributing factors include inadequate monitoring of vaccination records, limited public awareness of rabies prevention, and inefficiencies in healthcare service delivery (Cabrera et al., 2022).

At the local level, Animal Bite Treatment Centers (ABTCs) serve as primary facilities for rabies prevention and management. The typical workflow in ABTCs involves patient registration, assessment

of the animal bite case, administration of anti-rabies vaccines, scheduling of follow-up doses, and continuous patient monitoring. In Mandaluyong City, these centers often experience high patient volumes, especially during peak periods, which can strain available staff and resources (Mandaluyong City Health Office, 2023).

However, several challenges exist in the current workflow. Scheduling vaccinations is often done manually, making it difficult to organize patient appointments efficiently. This can result in overlapping schedules, missed appointments, and delays in service delivery. In addition, tracking follow-up doses is challenging because records are often paper-based, increasing the risk of incomplete or lost data (Luna & Reyes, 2022).

Another major concern is compliance with reporting requirements of the Department of Health (DOH). ABTCs are required to submit accurate and timely reports on vaccination records, patient data, and vaccine usage. Manual reporting processes are time-consuming and prone to errors, which may affect data accuracy and decision-making at the local and national levels.

Existing processes in many ABTCs rely heavily on manual systems, including paper-based registration, manual scheduling, and non-integrated inventory tracking. These practices lead to operational inefficiencies such as longer waiting times, inaccurate records, and difficulty in generating reports (Luna & Reyes, 2022).

Information systems can help address these challenges by automating data management, improving workflow efficiency, and enhancing the accuracy of reporting. Digital health systems have been shown to improve the efficiency and reliability of vaccination management processes (Nguyen et al., 2022).

Statement of the Problem

General Problem

The Mandaluyong City Animal Bite Treatment Center (ABTC) currently uses a manual and fragmented system for managing anti-rabies vaccination services. This results in inefficiencies in patient registration, scheduling, record-keeping, inventory monitoring, and reporting, which may affect service quality and patient care.

Specific Problems

1. Patient records, animal bite case details, and vaccination history are not efficiently managed, which may result in incomplete, inconsistent, duplicated, or difficult-to-retrieve information.
2. Patient flow and animal bite incident processing are not systematically monitored, particularly during peak consultation and vaccination periods, which may contribute to long waiting times, delayed service delivery, and difficulty tracking patient status.
3. Scheduling and monitoring of multi-dose anti-rabies vaccinations are not properly organized, making it challenging to track follow-up doses, identify patients with incomplete vaccination schedules, and ensure timely completion of the required doses.
4. Vaccine inventory, including stock levels, batch numbers, usage, and expiration dates, is not effectively monitored, which may lead to vaccine shortages, overstocking, expired vaccines, or wastage.
5. Patients may miss scheduled follow-up vaccinations due to the absence of an automated reminder mechanism, which can affect vaccination completion and continuity of treatment.
6. Preparation of vaccination-related reports for the Local Government Unit (LGU) and the Department of Health (DOH) is time-consuming, as data are manually compiled, summarized, and validated for compliance and monitoring purposes.
7. The quality of the developed system needs to be evaluated using ISO/IEC 25010:2011 software quality standards to determine whether it meets acceptable levels of functional suitability, usability, reliability, performance efficiency, security, compatibility, maintainability, and portability.
8. The acceptability of the developed system among Animal Bite Treatment Center staff needs to be assessed through User Acceptance Testing (UAT) to determine whether the system adequately supports patient registration, queue monitoring, vaccination scheduling, inventory tracking, SMS reminders, and report generation tasks.

Objectives of the Study

General Objectives

The general objective of this study is to design, develop, and evaluate a data-driven ticketing system for

the Mandaluyong City Animal Bite Treatment Center (ABTC) that will improve patient registration, queue management, vaccination scheduling, inventory monitoring, **automated notifications, and** reporting processes for anti-rabies vaccination management.

Specific Objectives

1. To develop a patient registration and records management module that enables the organized recording, storage, updating, and retrieval of patient profiles, animal bite case details, vaccination history, and related treatment records.
2. To implement a ticket queue management system that supports the monitoring of animal bite incidents, patient registration flow, service status, and patient processing to improve efficiency during peak consultation and vaccination periods.
3. To develop a vaccination scheduling and monitoring module that tracks multi-dose anti-rabies vaccination schedules, identifies patients with incomplete doses, and supports the proper organization of follow-up vaccination appointments.
4. To implement a vaccine inventory monitoring module that tracks vaccine stock levels, batch numbers, usage, and expiration dates to help prevent vaccine shortages, overstocking, and wastage.
5. To integrate an automated SMS reminder system that notifies patients of upcoming vaccination schedules and follow-up doses to help reduce missed appointments and improve vaccination completion rates.
6. To design a reporting dashboard that generates accurate, organized, and timely vaccination-related reports needed for Local Government Unit (LGU) and Department of Health (DOH) monitoring and compliance requirements.
7. To evaluate the developed system using ISO/IEC 25010:2011 software quality standards in terms of functional suitability, usability, reliability, performance efficiency, security, compatibility, maintainability, and portability.
8. To conduct User Acceptance Testing (UAT) among Animal Bite Treatment Center staff to assess the system's acceptability, usability, usefulness, and effectiveness in supporting patient registration, vaccination scheduling, queue monitoring, inventory tracking, SMS reminders, and report generation.

Scope and Limitation

This study focuses on the design and development of a data-driven ticketing system for the Mandaluyong City Animal Bite Treatment Center (ABTC). The system covers patient registration, ticket queue management, vaccination scheduling, inventory monitoring, automated SMS reminders, and reporting dashboard generation.

The proposed system will be developed as a web-based application accessible through computers and mobile browsers but will not include a dedicated mobile application. The development of the system will utilize modern web technologies, including HTML, CSS, JavaScript, PHP, and MySQL for database management. SMS notification functionality will also be integrated to support automated vaccination reminders.

The study is limited to the workflow of anti-rabies vaccination services and does not cover other services offered by the City Health Office. Implementation and pilot testing will be conducted only at the Mandaluyong City ABTC.

Significance of the Study

The proposed system will provide significant benefits to various stakeholders involved in anti-rabies vaccination management and public health services.

ABTC Staff and Healthcare Workers: Reduced manual workload, faster patient processing, accurate records, and real-time vaccine inventory monitoring will improve service delivery and reduce administrative burden.

Patients: Shorter waiting times, automated SMS reminders for follow-up doses, and better case monitoring will improve compliance and health outcomes.

Mandaluyong City Health Office and LGU: Accurate, automated reports will support faster decision-making and better resource allocation.

Department of Health (DOH): Standardized and timely data submission will improve national rabies surveillance and program evaluation.

Future Researchers: The study can serve as a reference for developing health information systems in other ABTCs or public health facilities.

Definition of Terms

Animal Bite Treatment Center (ABTC). A healthcare facility that provides treatment, vaccination, and monitoring for animal bite patients.

Data-Driven Ticketing System. A digital system designed to manage patient flow, scheduling, and data processing using organized and automated tools.

Department of Health (DOH). The national government agency responsible for public health services and policies in the Philippines.

Local Government Unit (LGU). The local authority responsible for implementing health programs and services within a specific area.

Post-Exposure Prophylaxis (PEP). A treatment given after exposure to rabies to prevent infection.

CHAPTER 2

Literature Review

The chapter is a comprehensive review of the scholarly materials and information on data-driven ticketing systems, rabies vaccination management, queuing optimization, spatial analysis in public health, decision support systems, software quality standards, and user acceptance testing in healthcare operations. This chapter also discusses the application and implications of these technologies in the efficiency of the vaccination management, areas for further research, and implications for local government units and animal health services. This chapter also presents the rationale for the design, development, and testing of the data-driven ticketing system for anti-rabies vaccination management based on the review of the related literature.

I. Overview

Rabies Vaccination Management and Public Health Impact

Rabies is an important public health concern that persists as an endemic disease in various parts of the world. Developing countries are the most affected by dog-mediated rabies. Vaccination campaigns against rabies have been the most important strategy for controlling and eliminating the disease. The WHO aims to achieve zero human rabies deaths by the year 2030. A study conducted by Ferguson et al. (2025) and Sambo et al. (2024) showed that the success of dog vaccination campaigns against rabies is influenced by the need to attain and maintain high vaccination coverage rates and minimize the spatial heterogeneity of the vaccination coverage.

In the Philippines, the control of rabies is considered a priority. A study on the in-depth analysis of the incidence of human rabies in Metro Manila from 2006 to 2015 was conducted by Guzman et al. (2022). It was established that the incidence of rabies has spatial clusters. Therefore, the areas that were considered as high- risk areas were identified. Another study was conducted by Chaudhari et al. (2022), which was on the ecology and dog demographics of some areas in the Philippines. It was established that it was important to identify the dynamics of the dog population in the Philippines so that the vaccination

program could be conducted. Dizon et al. (2022) also contributed to the field of rabies research by conducting a household survey in Bulacan.

Furthermore, Bandoy et al. (2025) also presented a study on the government-run animal bite treatment center in Iloilo City and its functions and challenges in providing rabies post-exposure prophylaxis. The study is a descriptive study on its sixth year. The result of this study is reflective of the systemic problems experienced by patients in terms of long hours of waiting and documentation.

Queuing Systems and Waiting Time Optimization

The relationship between waiting time and patient satisfaction is well-established in the literature. Yurizali & Adhyka (2022) used a structural equation modeling approach in their research on patient satisfaction in relation to waiting time and fast-track service provision. It was found that waiting time impacts patient satisfaction levels in a negative manner and may deter patients from seeking further treatment. This may have particular relevance in the context of vaccine administration programs where repeat doses may be necessary.

Anuruddhika et al. (2022) carried out a systematic literature review on the approaches in queuing modeling. According to them, there are different mathematical and computational techniques for optimization in the delivery of services in situations where demand is variable and resources are scarce. The literature review on the application of queuing theory in healthcare services with stochastic arrivals and varying service times is also noteworthy. Queuing theory may be applied in developing a plan for a campaign on a rabies vaccination program with a surge in demand during scheduled events.

Although Haghi et al. (2022) were primarily interested in drug delivery systems, the authors also discussed the idea of a "smart scheduling and controlled release," which can metaphorically be related to vaccination campaigns. Haghi et al. have also discussed the idea of using poly(l-histidine) grafted carbon nanotubes as "smart drug delivery systems."

Data-Driven Decision Support in Healthcare

The incorporation of data analytics in the healthcare decision-making process has transformed the administration of vaccine programs. Batoon & Piad (2023) have proposed a web-based scheduling system with geo-tagging integration and decision support for health centers in the region. This shows the potential of using technology in improving vaccine programs based on demand and supply. It incorporates the supply of vaccine, patient registration, and geographical distribution.

The use of machine learning techniques in healthcare prediction and decision support has gained momentum in recent times. Bukhari et al. (2022) used decision tree-based ensemble machine learning models in predicting Zika virus T-cell epitopes as vaccine candidates. This demonstrates the use of algorithmic techniques in vaccine development and targeting. Kalegele and Lubua (2025) used decision tree classifiers in predicting factors associated with childhood vaccination uptake. This demonstrates the use of machine learning techniques in understanding vaccination behaviors. Battista et al. (2023) used decision trees in population health surveillance research.

Arciniegas et al. (2025) employed a hybrid approach that incorporated Markov models and decision trees to estimate the public health and economic implications of vaccination strategies in Colombia. The hybrid modeling approach employed by Arciniegas et al. is a suitable example of a decision support system that can be employed to assess different vaccination strategies.

Tuan et al. (2025) developed personalized machine learning models that can predict missed appointments in primary care services. The models developed by Tuan et al. are directly applicable to vaccination strategies that are affected by missed appointments. Osorio et al. (2025) improved upon Tuan et al.'s work by developing a prediction model for missed appointments at outpatient services using different machine learning models. The accuracy of Osorio et al.'s models is suitable for implementing proactive strategies to prevent missed appointments.

Gupta and Gupta (2025) studied the supply chain management of pharmaceuticals. They emphasized the role of customer service and the delivery of medicine in the case of shortage. The principles of the research could be directly applied to vaccine supply chain management in the control of rabies.

Spatial Analysis and Geographic Information Systems in Rabies Control

Geographic information systems have emerged as essential tools in rabies surveillance and the management of vaccine campaigns. In a series of studies, Ferguson et al. (2024, 2025) have attempted to highlight the significance of the minimization of spatial heterogeneity in the distribution of vaccines by demonstrating how it impacts herd immunity as well as the maintenance of persistence. The studies are a basis for advocating for more campaigns in areas that are underserved as highlighted by the spatial mapping.

Chandran and Roy (2024) undertook a study in which they undertook a systematic review of studies that had utilized the application of geographic information systems in health research in India.

The authors were able to highlight the application of spatial analysis in the identification of diseases, health services planning, and evaluation. The study is methodologically significant in the management of rabies because it highlights the need for the maximization of vaccines.

In a research conducted by Zakharova and Liskova (2024) on the pattern of animal rabies in Russia, spatial analysis was used in the research to identify the factors associated with the risk of the disease. It is a clear indication of how epidemiological research can be used in conjunction with geographic information systems in understanding the environment and factors associated with the disease.

In comparative research conducted by Eugene et al. (2025) on a comparative study of knowledge of rabies and practices of prevention in Nigerian hotspots and non-hotspot areas using spatial analysis, it is evident that geographic differences are a major area of concern.

Wambugu et al. (2024) conducted research on rabies virus geographic distribution in Kenya. This research was conducted using a combination of phylogenetics and spatial data, and it provided information on rabies virus transmission dynamics and areas that might require vaccine strain adaptation.

In another research study, Xie et al. (2025) conducted research on the association between neighborhood socioeconomic status and dog-mediated rabies in Latin America. This research study showed that there are differences in rabies incidence and surveillance levels. This research was conducted using spatial analysis, and it was shown that rabies is more common in low socioeconomic areas and that surveillance is low in those areas.

Philippine-Specific Rabies Research

Several researches have also explored the epidemiology of rabies. These researches provided significant background materials for the proposed system. The research of Satur et al. (2025) employed the use of optimal control theory in the epidemiological modeling of rabies in Davao City. Time-dependent vaccination strategies were also incorporated into the research. Although it is not exactly about simulating an algorithm that uses scheduling as its main concept, the mathematical model presented in the research provides significant background materials about the role of time-dependent factors in the epidemiology of rabies.

The most comprehensive research about the epidemiology of human rabies in Metro Manila was conducted by Guzman et al. (2022). The study by Chaudhari et al. (2022) provided critical information

on the ecology and demographics of dogs in the Philippines. Their study revealed that the population of owned dogs in the country varied significantly. It also revealed that the population of free-roaming dogs affected the estimation of the number of vaccinated dogs. These are critical findings that impact the vaccination drive significantly.

The study by Dizon et al. (2022) evaluated the knowledge of the public on rabies and the ownership of dogs in the province of Bulacan. It revealed that the public had a poor knowledge of the transmission and prevention of rabies. These findings are critical for the development of a system that incorporates vaccination and education on the prevention of rabies.

The study by Bandoy et al. (2025) evaluated the operational challenges faced by a government-run animal bite treatment center. These challenges include the documentation of clients, waiting time for clients, and the lack of resources. These are critical findings that impact the development of the proposed system.

Software Quality Standards: ISO/IEC 25010

The quality of healthcare information systems is of great importance in ensuring quality and timely service is provided in vaccine-related applications. ISO/IEC 25010 standards define that software quality can be described through various quality characteristics such as functional suitability, reliability, usability, efficiency in performance, maintainability, and security. Mukhlis et al. (2023) used the Model View Controller method in designing an information system for pet care services, hence ensuring software quality and maintainability.

If the standards and guidelines provided in ISO/IEC 25010 are considered in the proposed information system, then it can be guaranteed that even if there are limited IT resources available in local government units, then high-quality service can be provided to customers through reliability, usability, and efficiency. This will help improve efficiency in operations by ensuring that no possible error is experienced in vaccine tracking, hence ensuring customer satisfaction. These standards and guidelines can be used to improve or expand the information system to cater to more barangays or rabies elimination programs.

User Acceptance Testing (UAT) in Healthcare Systems

User Acceptance Testing is conducted to ascertain that healthcare information systems meet the needs and expectations of the users. In the case of rabies vaccination programs, the use of User

Acceptance Testing is significant because it will ensure that the users of the healthcare information system, including veterinary staff and barangay health workers and administration personnel, will be able to use the system more efficiently.

The feedback gathered during the testing will also allow the healthcare information system developers to refine features of the system that will be of great use to the users. The use of ISO/IEC 25010 will allow the healthcare information system to be both efficient and effective.

II. Gaps in Knowledge and Practice

Integrated Ticketing System for Animal Bite Treatment Centers: While queuing and appointment systems have been extensively studied in general healthcare settings, there is a notable absence of research on integrated ticketing systems specifically designed for Animal Bite Treatment Centers (ABTCs). Existing studies do not address how ticketing workflows can be optimized to manage high patient volumes, prioritize bite severity cases, and reduce waiting times in the ABTC context.

Data-Driven Queue Management in Public Health Facilities: Most studies on queue management focus on hospitals and general outpatient clinics. There is limited research on how data-driven ticketing systems can be applied in local government-operated health facilities such as ABTCs, particularly in managing patient flow during peak periods and emergency walk-in cases.

Digital Vaccination Scheduling and Dose Monitoring for PEP Compliance: Existing literature lacks focus on automated scheduling systems tailored for Post-Exposure Prophylaxis (PEP) protocols. Manual tracking of multi-dose vaccination schedules increases the risk of missed follow-up doses, yet no integrated digital solution has been developed specifically for ABTC operations.

Automated Client Notification Systems in Animal Bite Management: While reminder systems have been studied in general healthcare appointments, there is insufficient research on the integration of automated SMS notification features within animal bite treatment workflows to reduce missed vaccination appointments and improve PEP completion rates.

Real-Time Vaccine Inventory Monitoring in LGU Health Facilities: Studies on vaccine inventory management are largely focused on national or regional supply chains. There is a gap in

research addressing real-time, system-level inventory tracking at the individual ABTC level, particularly for monitoring stock levels, usage rates, and expiration dates of anti-rabies vaccines. Software Quality Evaluation of Health Information Systems in LGU-Operated Facilities: While ISO/IEC 25010 standards provide a comprehensive framework for software quality evaluation, there is limited research applying these standards to health information systems developed for local government-operated facilities. This gap highlights the need for systematic evaluation of system reliability, usability, and performance in resource-constrained settings such as ABTCs.

Automated Reporting and DOH/LGU Compliance in ABTC Operations: Existing research does not adequately address the challenge of generating timely, accurate, and standardized reports for DOH and LGU compliance in the context of animal bite treatment. Manual reporting processes remain a significant operational burden, and no data-driven reporting dashboard tailored for ABTC compliance requirements has been documented in the literature.

III. Relevance to the Proposed Project

The reviewed literature highly supports the need for an integrated data-driven ticketing system for anti-rabies vaccination management. Literature on the epidemiology of rabies in the Philippines was reviewed, and these studies by Guzman et al. (2022), Chaudhari et al. (2022), Dizon et al. (2022), and Satur et al. (2025) established the importance of optimizing anti-rabies vaccination management and identified some specific challenges in the local context.

Literature on the application of queuing systems and waiting time optimization was reviewed, and these studies by Yurizali & Adhyka (2022) and Anuruddhika et al. (2022) established a relationship between waiting times and customer satisfaction with the services provided, thus emphasizing the importance of optimizing waiting times in encouraging pet owners to participate in anti-rabies vaccination management.

The existing research on decision support systems and machine learning in healthcare environments (Batoon & Piad, 2023; Kalegele & Lubua, 2025; Tuan et al., 2025; Osorio et al., 2025) is applicable to the data analytics part of the proposed system.

The existing research on spatial analysis (Ferguson et al., 2024, 2025; Chandran & Roy, 2024;

Xie et al., 2025) is applicable to the proposed system, particularly in terms of methodological insights related to the proposed system's geo-integration part.

The insights from the ISO/IEC 25010 standard (Mukhlis et al., 2023) are applicable to the proposed system, particularly in terms of the importance of designing a system that is grounded on existing software quality standards.

The insights from the User Acceptance Testing part of the proposed system validate that not only is the proposed system effective from a technical point of view, but it is also consistent with users' expectations in veterinary public health environments.

Lastly, the operational challenges that are encountered at animal bite treatment centers in the Philippines, as mentioned by Bandoy et al. (2025), which are already existing challenges, are a motivation for this proposed system. The challenges are related to documentation and waiting time at rabies prevention services.

IV. Conceptual Framework

The conceptual framework provides an overview of the essential components and their relationships in the development and assessment of the proposed data- driven ticketing system in the management of the anti-rabies vaccination. The proposed conceptual framework provides an overview of the relationships between ticket optimization, spatial data integration, and client notification systems. The proposed conceptual framework also provides an overview of the importance of the proposed ticketing system based on the use of software quality standards (ISO/IEC 25010) and client acceptance testing. The proposed conceptual framework provides a clear visual representation of the proposed ticketing system and how it would be useful in improving the efficiency of the operations of the vaccination campaign.

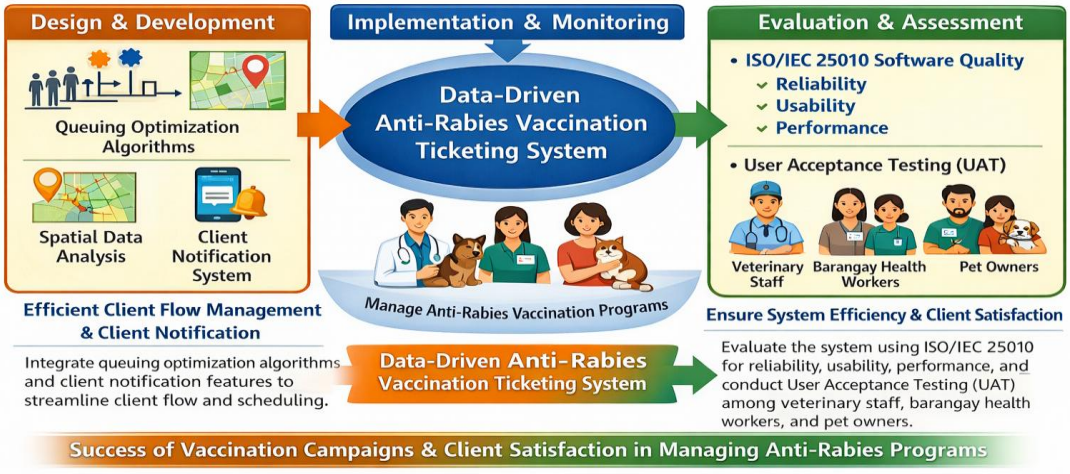


Figure 1.0 Conceptual Framework of the Data-Driven Ticketing System for Optimized Anti-Rabies Vaccination Management

The conceptual framework of the study is based on the Input-Process-Output (IPO) Framework, which serves as the guide in the development of the proposed data-driven ticketing system for anti-rabies vaccination management. The Input component includes the resources, requirements, patient information, vaccination records, inventory data, and system requirements needed for the study. The Process component involves the design, development, implementation, and testing of the system, including patient registration, queue management, vaccination scheduling, inventory monitoring, SMS notification, and report generation. Lastly, the Output component represents the expected results of the study, which include an efficient and organized ticketing system, improved vaccination monitoring, accurate reporting, reduced waiting time, and enhanced healthcare service delivery at the Mandaluyong City ABTC. The framework also illustrates the implementation and evaluation of the proposed system.

The importance of integrating queuing optimization techniques and client notification is shown in the design process of the project. After the implementation of the ticketing system, there is an evaluation process to ensure that the ticketing system is working efficiently in a real-world environment. The ISO/IEC 25010 standard is used in evaluating the reliability, usability, and performance of the ticketing system. The User Acceptance Testing is used in measuring client satisfaction and acceptance

of the ticketing system among the veterinary staff, barangay health workers, and pet owners. The significance of client flow management is shown in ensuring the success of a vaccination campaign and client satisfaction in the implementation of a ticketing system in managing anti-rabies vaccination programs.

Synthesis

Data-driven methods have been shown to be effective drivers of efficiency in healthcare service delivery, including vaccination services. For instance, rabies epidemiological studies in the Philippines show that high and equitable coverage is critical to rabies elimination. However, systemic barriers such as long waiting times, documentation requirements, and geographical inequities hinder service delivery (Guzman et al., 2022; Chaudhari et al., 2022; Dizon et al., 2022). Queuing systems literature shows that efficient management of waiting times is critical to client satisfaction and service utilization (Yurizali & Adhyka, 2022; Anuruddhika et al., 2022).

Decision support systems and machine learning technologies provide opportunities for predicting vaccine uptake, identifying determinants of non-appointment attendance, and optimizing resource allocation (Batoon & Piad, 2023; Kalegele & Lubua, 2025; Tuan et al., 2025; Osorio et al., 2025). These technologies provide opportunities for a proactive approach to vaccine management as opposed to a reactive approach to vaccine demand.

Aside from that, software quality and user acceptance are also important factors that are necessary for ensuring that a data-driven ticketing system is a success. According to the ISO/IEC 25010 standard, reliability, user experience, and efficiency are necessary for ensuring that a data-driven ticketing system meets the industry standard (Mukhlis et al., 2023). On the other hand, user acceptance test is necessary for ensuring that a data-driven ticketing system meets user requirements and expectations. The test is conducted in a real world environment. These are important factors that show that a data-driven ticketing system that is a success is not only effective in its operations but also meets software quality and user acceptance for a successful vaccination campaign.

To give a background and show the contribution of previous studies in the development and validation of the proposed data-driven ticketing system in the management of anti-rabies vaccination, the following table presents some of the important studies in support of the proposed concept.

TABLE 1.0 MATRIX OF RELATED LITERATURE

No	Title	Author(s)	Year	Source	Type	Contribution
1	Development of the poly(l-histidine) grafted carbon nanotube as a possible smart drug delivery vehicle	Haghi, A., Raissi, H., Hashemzadeh, H., & Farzad, F.	2022	Computers in Biology and Medicine	International	Introduces smart scheduling concepts applicable to adaptive vaccination campaign management
2	Optimizing Vaccine Access: A Web-Based Scheduling System with Geo-Tagging Integration and Decision Support for Local Health Centers	Batoon, J., & Piad, K.	2023	Open Journal of Applied Sciences	International	Demonstrates web-based scheduling with geo-tagging for local health centers; directly relevant to proposed system
3	The Relationship of Patient Satisfaction with the Waiting Time and Fast Track Services: Structural Equation Modelling Test	Yurizali, B., & Adhyka, N.	2022	Advances in Health Sciences Research	International	Establishes relationship between waiting times and patient satisfaction; supports need for ticketing optimization

4	The Approaches Utilized in Queuing Modeling: A Systematic Literature Review	Anuruddhika, T., Prasanth, S., & Rathnayaka, R.	2022	Asian Journal of Convergence in Technology	International	Reviews queuing models applicable to healthcare service optimization
5	Pharmaceutical Supply Chain Role in Managing Customer Service and Medicine Delivery amid Shortages to USA Customers	Gupta, G., & Gupta, M.	2025	International Journal of Multidisciplinary Evolutionary Research	International	Examines supply chain management principles applicable to vaccine distribution
6	Estimating the potential public health and economic impact of vaccination strategies with an adapted vaccine in Colombia	Arciniegas, J., et al.	2025	Expert Review of Vaccines	International	Demonstrates combined Markov-decision tree modeling for vaccination strategy evaluation
7	Decision tree based ensemble machine learning model for the prediction of Zika virus T-cell epitopes	Bukhari, S., Webber, J., & Mehbodniya, A.	2022	Scientific Reports	International	Illustrates machine learning applications in vaccine development and targeting

8	Determinants of Childhood Vaccination Uptake: A Machine Learning Approach Using a Decision Tree Classifier	Kalegele, K., & Lubua, E.	2025	The Journal of Informatics	International	Identifies factors influencing vaccination uptake using decision tree analysis
9	Examining the use of decision trees in population health surveillance research	Battista, K., et al.	2023	Health Promotion and Chronic Disease Prevention in Canada	International	Demonstrates decision tree applications in population health data analysis
10	Improved effectiveness of vaccination campaigns against rabies by reducing spatial heterogeneity in coverage	Ferguson, E., et al.	2025	PLOS Biology	International	Shows importance of spatially uniform coverage for rabies vaccination effectiveness
11	Understanding and overcoming geographical barriers for scaling up dog vaccinations against rabies	Sambo, M., et al.	2024	Scientific Reports	International	Identifies geographic barriers to dog vaccination and strategies to address them

12	Rabies knowledge and prevention practices in Gombe state, Nigeria: a community-based comparative cross-sectional study	Eugene, E., et al.	2025	BMC Public Health	International	Documents geographic disparities in rabies knowledge and vaccination practices
13	Geographic Distribution of Rabies Virus and Genomic Sequence Alignment of Wild and Vaccine Strains, Kenya	Wambugu, E., et al.	2024	Emerging Infectious Diseases	International	Combines phylogenetic and spatial analysis for rabies surveillance
14	Reducing spatial heterogeneity in coverage improves the effectiveness of dog vaccination campaigns against rabies	Ferguson, E., et al.	2024	bioRxiv	International	Preprint demonstrating spatial analysis methods for vaccination campaign optimization

15	Applications of geographical information system and spatial analysis in Indian health research: a systematic review	Chandran, A., & Roy, P.	2024	BMC Health Services Research	International	Systematic review of GIS applications in health research; methodological guidance
16	Patterns of animal rabies in the Nizhny Novgorod region of Russia (2012–2022): the analysis of risk factors	Zakharova, O., & Liskova, E.	2024	Frontiers in Veterinary Science	International	Identifies spatial risk factors for animal rabies using geographic analysis
17	Neighborhood Socioeconomic Status and Dog-Mediated Rabies: Disparities in Incidence and Surveillance Effort	Xie, S., et al.	2025	medRxiv	International	Reveals socioeconomic disparities in rabies burden and surveillance
18	Clinical, epidemiological, and spatial features of human rabies cases in Metro Manila, the Philippines	Guzman, F., et al.	2022	PLoS Neglected Tropical Diseases	Local	Comprehensive analysis of Philippine rabies cases; identifies high-risk areas

	from 2006 to 2015					
19	Dog Ecology and Demographics in Several Areas in the Philippines and Its Application to Anti-Rabies Vaccination Programs	Chaudhari, A., et al.	2022	Animals	Local	Provides essential data on Philippine dog populations for vaccination planning
20	Household survey on owned dog population and rabies knowledge in selected municipalities in Bulacan, Philippines	Dizon, T., et al.	2022	PLoS Neglected Tropical Diseases	Local	Assesses dog ownership patterns and rabies knowledge in Philippine communities
21	Optimal Control Theory Applied to Rabies Epidemiological Model with Time-dependent	Satur, D., et al.	2025	Acta Medica Philippina	Local	Mathematical modeling of vaccination timing; supports data-driven scheduling

	Vaccination in Davao City					
22	Predicting Missed Appointments in Primary Care: A Personalized Machine Learning Approach	Tuan, W., et al.	2025	Annals of Family Medicine	International	Predicts no-show appointments; applicable to vaccination appointment management
23	Predicting no-shows at outpatient appointments in internal medicine using machine learning models	Osorio, F., et al.	2025	PeerJ Computer Science	International	Demonstrates machine learning for no-show prediction; informs reminder system design
24	Model View Controller Method For Animal Care (Petcare) Information System At Niz Petcare Lawang	Mukhlis, I., Hermansyah, D., & Hariyanti, T.	2023	JEECS	International	Demonstrates software architecture for pet care information systems

25	The struggle in the attempt to eliminate dog-mediated human rabies in Nigeria by 2030: a retrospective study	Vakuru, C., & Kwaghe, A.	2024	PAMJ-One Health	International	Documents challenges in rabies elimination efforts in resource-limited settings
26	Dog-Mediated Rabies Surveillance in Nigeria (2014–2023): Investigating Seasonality and Spatial Clustering	Williams, R., et al.	2025	Tropical Medicine and Infectious Disease	International	Analyzes spatial and temporal patterns in rabies surveillance data
27	Where rabies is not a disease. Bridging healthworlds to improve mutual understanding and prevention of rabies	Nadal, D., et al.	2022	Frontiers in Veterinary Science	International	Examines cultural factors affecting rabies prevention; informs community engagement
28	A cross-sectional study on antirabies vaccination exercise commemorating the World Rabies Day	Muhinda, O. V., et al.	2025	PAMJ - One Health	International	Documents vaccination campaign operations; identifies practical challenges

	Celebration 2024 in Plateau State, Nigeria					
29	Functions and challenges of a government-run animal bite treatment center in the delivery of rabies post-exposure prophylaxis services in Iloilo City, Philippines	Bandoy, A. B., et al.	2025	Dialogues in Health	Local	Documents operational challenges in Philippine rabies treatment centers; directly motivates proposed system
30	Rabies-preventive health program for mothers in maternal and child health centers: quasi experimental study	Baraka, S. I. M., et al.	2025	Journal of Health Population and Nutrition	International	Evaluates rabies education interventions; informs client communication design

Chapter 3

Methodology

This chapter presents the research design, software development methodology, system design and development, analytics methodology, data collection procedures, and statistical treatment of data used in the development and evaluation of N-VISION: A Data-Driven Ticketing System for Optimized Anti-Rabies Vaccination Management for the Mandaluyong City Animal Bite Treatment Center (ABTC).

The chapter is organized into the following sections: (1) Research Design, which describes the type of research approach used; (2) Research Procedure, which outlines the step-by-step process followed in conducting the study; (3) Software Development Methodology, which explains the Agile Scrum framework used in building the system; (4) System Design and Development, which details the architecture, modules, and technical specifications of the proposed system; (5) Analytics Methodology, which describes the data analytics approach used in processing and presenting operational data through the reporting dashboard; (6) Data Collection, which describes the instruments and respondents used in evaluating the system; and (7) Statistical Treatment of Data, which explains the descriptive statistical methods applied in analyzing the evaluation results.

Research Design

This study employs a quantitative research design integrating developmental research with a descriptive- evaluative approach. This design is appropriate because it allows for both the systematic creation of a functional information system and the structured measurement of its quality, performance, and user acceptance using numerical data.

The study is grounded in developmental research, a systematic approach that involves the design, development, and evaluation of practical solutions to real-world problems. Developmental research typically engages in cyclic construction and testing of artifacts to progressively refine their effectiveness and applicability (Herlandy & Syahfutra,2023). In this study, the artifact being developed is a data-driven ticketing system that addresses identified operational challenges at the Mandaluyong City Animal Bite Treatment Center (ABTC).

Within this framework, a descriptive-evaluative approach is applied. Descriptive research involves observing and documenting phenomena as they naturally occur without manipulating variables,

thereby providing a detailed account of system characteristics as experienced by users (Creswell & Guetterman, 2022). The evaluative component allows the researchers to measure how well the system meets predefined quality and usability criteria. Evaluative research is widely used in system assessment studies to determine effectiveness, reliability, security, and user satisfaction through structured indicators (Bryman, 2022).

Quantitative data are collected through structured survey questionnaires using a 4-point Likert scale and analyzed using descriptive statistics to assess system quality and user acceptance. This approach is consistent with ISO/IEC 25010 software quality standards, which define measurable quality characteristics including functional suitability, reliability, performance efficiency, usability, and security (ISO/IEC, 2023).

Research Procedure

The research procedure follows a systematic, multi-stage process to ensure that the development and evaluation of the Data-Driven Ticketing System for Anti-Rabies Vaccination Management is traceable, rigorous, and explicitly aligned with the study's specific objectives.

Problem Definition and Objective Setting

The procedure begins with a clear articulation of the operational challenges experienced by the Mandaluyong City ABTC, as presented in Chapters I and II. These challenges include manual patient registration, inefficient vaccine scheduling, poor inventory tracking, high patient waiting times, difficulty in monitoring incomplete vaccination schedules, and time-consuming reporting to the DOH and LGU. These identified issues were translated into specific system objectives focused on the digitization and automation of ABTC operations.

Mapping of Research Procedure to the Study Objectives

To ensure full methodological alignment, each stage of the research procedure is explicitly linked to the study objectives as presented in the table below:

Table 2.1 A Data-Driven Ticketing and Operational Automation System for ABTC

Research Procedure Stage	Study Objectives Supported	Description
Literature Review and Conceptual Framework Development	Objectives 1–4	Provides theoretical foundation for system design, evaluation standards, and user acceptance
Requirements Analysis and System Specification	Objectives 1–4	System functions are derived from ABTC operational problems including registration, scheduling, and communication gaps
System Design and Prototyping	Objectives 1–3	Focuses on development of core modules such as patient registration, ticket queue management, scheduling, and inventory monitoring
System Development and Internal Testing	Objectives 1–4	Ensures system functionality, integration, and readiness for evaluation based on ISO/IEC 25010 standards
Pilot Deployment and Data Collection	Objective 4	Actual system performance, usability, and user acceptance are evaluated through feedback
Data Analysis and Interpretation	Objective 4	Descriptive statistics are used to interpret system evaluation and user acceptance results
Revision and Documentation	Objective 4	Incorporates User Acceptance Testing (UAT) feedback for system improvement

Gantt Chart for the Development of the Data-Driven Ticketing System

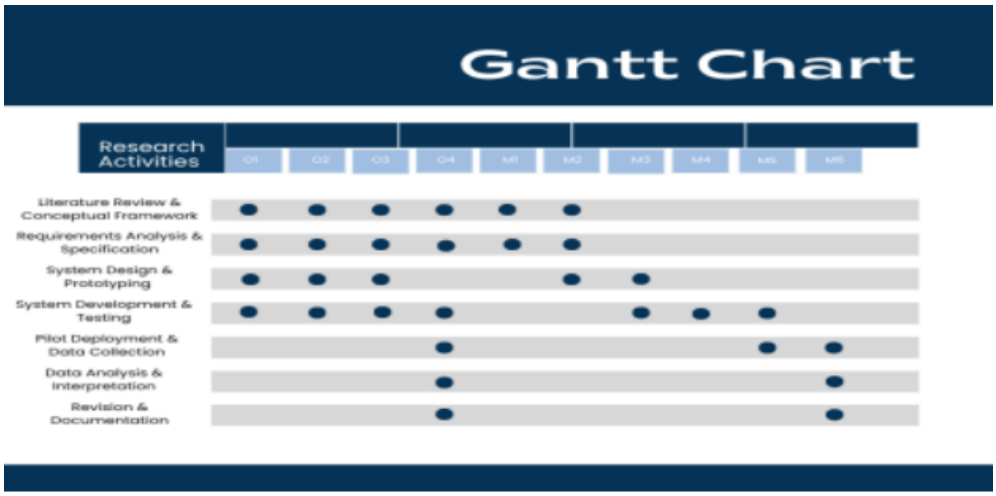


Figure 3.0 Gantt Chart of Research Procedure Aligned with study Objectives

Literature Review and Conceptual Framework Development

A comprehensive review of local and international literature pertaining to rabies epidemiology, queuing theory, decision support systems, spatial analysis, software quality standards, and user acceptance models informed the conceptual framework of the study. The results of this review are presented in Chapter II and guided the development of the system, which adheres to the Input-Process-Output-Outcomes (IPOO) model to ensure alignment with best practices in health informatics.

Requirements Analysis and System Specification

Functional requirements encompass patient registration, queue management, vaccination scheduling, inventory tracking, automated communication, and reporting functionalities. Non-functional requirements are derived from ISO/IEC 25010 quality attributes, emphasizing system performance, security, usability, and reliability.

System Design and Prototyping

Using the requirements specifications, the researchers designed the system architecture, database schema, and user interface wireframes aligned with the actual workflows of the Mandaluyong City ABTC. An initial prototype was developed and iteratively refined through consultations with veterinary staff and barangay health workers to ensure that the system accurately reflects user roles, operational constraints, and existing service delivery processes.

System Development and Internal Testing

Applying Agile Scrum methodologies, development is organized into incremental sprints, each delivering validated modules. Systematic unit and integration testing are conducted to verify compliance with functional specifications prior to deployment.

Pilot Deployment and Data Collection

The system was developed using the Agile Scrum methodology, with work organized into iterative sprints for each major module. During each sprint, module-level and integration-level testing were conducted to validate core functionalities, including patient registration, ticket generation, vaccination scheduling, inventory updates, automated SMS notifications, and report generation, prior to deployment.

Data Analysis and Interpretation

Survey responses were coded and analyzed using descriptive statistical measures, including frequency distribution, percentage, and weighted mean, to evaluate system quality and user acceptance in relation to the study's specific objectives. The findings were interpreted in the context of the research questions, study objectives, and related literature presented in Chapter II.

Revision and Documentation

Feedback from pilot deployment informs iterative refinements. Comprehensive documentation encompasses development decisions, testing outcomes, evaluation results, and recommendations for future enhancements and scalability considerations.

Software Development Methodology

The development of the Data-Driven Ticketing System for Anti-Rabies Vaccination Management follows the Agile software development methodology, a flexible and iterative approach that emphasizes collaboration, continuous improvement, and the incremental delivery of working software (Atlassian, 2026). This methodology is well-suited to the present study because the requirements of the Mandaluyong City ABTC may evolve during the development process, and frequent feedback from ABTC staff is necessary to ensure the system remains aligned with actual operational workflows.

Agile development is characterized by short iterative cycles that allow the development team to

continuously refine and improve the system based on user feedback. Each iteration delivers a working increment of the system, which is reviewed and evaluated before proceeding to the next cycle. This approach ensures that the system is developed in close collaboration with ABTC stakeholders, allowing for immediate adjustments to features and functionalities as needed.

For this study, the development is structured into six iterations, each two weeks in duration, as summarized in Table 3.1. Each iteration focuses on a specific set of system modules and deliverables, ensuring that the system is developed in a systematic and organized manner while remaining responsive to the needs of ABTC personnel.

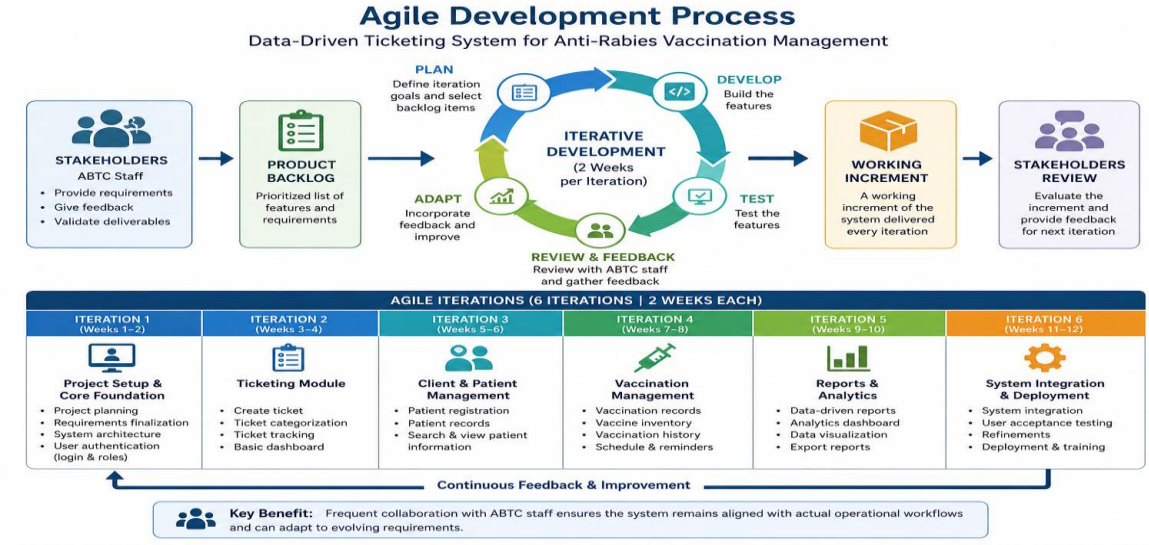


Figure 4.0 Agile Development Process

Table 3.0. Iteration Plan for the Data-Driven Ticketing System Development

Interation	Duration	Focus Area	Key Deliverables
Interation 1	2 Weeks	Requirements & System Setup	Requirements Gathering, System Architecture Design, Development Environment Setup
Interation 2	2 Weeks	Patient Registration & Ticket Queue	Patient Registration Module, Ticket Queue System, Queue

			Management Features
Interation 3	2 Weeks	Vaccination Scheduling & Dose Monitoring	Vaccination Scheduling Module, Dose Monitoring & Tracking, Reminder Schedule Setup
Interation 4	2 Weeks	Vaccine Inventory & SMS Reminders	Vaccine Inventory Management, SMS Reminder Integration, Notification Settings
Interation 5	2 Weeks	Reporting Dashboard & Admin Features	Reporting Dashboard, Data Analytics & Visualization, Admin Management Features
Interation 6	2 Weeks	Testing, UAT, and Deployment	System Testing (Unit, Integration, System), User Acceptance Testing (UAT), Deployment & Go-Live

Data Collection

Data for this study are collected through survey questionnaires and system log review during the pilot deployment and early operational use of the Data-Driven Ticketing System at the Mandaluyong City ABTC. This approach ensures that structured, comprehensive feedback is gathered from both technical experts and ABTC end users regarding the system's technical performance and practical usability.

Two research instruments are utilized in this study:

ISO/IEC 25010 Evaluation Instrument

This instrument is based on the ISO/IEC 25010:2011 software quality model and evaluates the system against five quality characteristics: functional suitability, performance efficiency, usability, reliability, and security. The instrument ensures alignment with internationally recognized software quality standards and provides a rigorous benchmark for

system evaluation (ISO/IEC, 2023; Mukhlis et al., 2023).

User Acceptance Testing (UAT) Instrument

A self-developed questionnaire is used to assess user acceptance, ease of use, satisfaction, and intention to continue using the system among ABTC personnel. This instrument captures acceptance factors specific to the Mandaluyong City ABTC that are not fully addressed by technical quality standards alone.

Both instruments employ a 4-point Likert scale with satisfaction-oriented verbal interpretation. The scale intentionally excludes a neutral midpoint to encourage respondents to take a definitive stance, thereby minimizing central tendency bias and generating decisive, actionable feedback (Kusmaryono et al., 2022; Ramu, 2023).

Table 4.0 – 4-Point Likert Scale with Verbal Interpretation for System Evaluation and User Acceptance

Scale Value	Description	Mean Range	Verbal Interpretation
4	Strongly Agree	3.26 - 4.00	Highly Satisfied
3	Agree	2.51 - 3.25	Satisfied
2	Disagree	1.76 - 2.50	Dissatisfied
1	Strongly Disagree	1.00 - 1.75	Highly Dissatisfied

Both instruments undergo reliability and validity testing prior to full administration. Cronbach's alpha is used to measure internal consistency, with values of 0.70 or higher considered acceptable and 0.80 or higher indicating good reliability (Taherdoost, 2022). Content validity is ensured through expert review and pilot testing to confirm that each item accurately measures its intended construct.

Respondents Segmentation

The study involved a total of eight (8) respondents, composed of two (2) technical experts and six (6) operational end-users from the Mandaluyong City Animal Bite Treatment Center (ABTC). Given the specialized nature of the system evaluation, the study employed a small but targeted population representing all key user roles involved in ABTC operations.

- The technical experts (2 IT professionals/system evaluators) were selected through purposive sampling, as their expertise in information systems and software evaluation is necessary for assessing compliance with ISO/IEC 25010 standards.
- The operational end-users (6 staff members) consisting of 2 nurses, 2 doctors, and 2 administrative staff (encoders) were selected through total enumeration, since all available personnel in these categories were included due to the limited population size of the ABTC.

This sampling approach is appropriate for system development and evaluation studies, where the focus is on expert judgment and actual system users rather than population generalization. Specifically, the study employed purposive sampling for the technical experts, as their expertise in information systems and software evaluation is necessary for assessing compliance with ISO/IEC 25010 standards. Meanwhile, total enumeration was used for the operational end-users, since all available personnel in the categories of nurses, doctors, and administrative staff were included due to the limited population size of the ABTC.

Table 5.0 Respondents Profile

Respondents Group	**Role in Ticketing Systems**	**n**
IT Support Staff	Ticketing Resolvers/System Admins	2
Nurse	Tickets Requesters - Clinic	2
Doctors	Tickets Requesters - Clinic	2
Administrative Staff	Tickets Requesters - Non Clinic	2
Total		8

Table 6.0 Descriptive Statistical Techniques Applied to Study Objectives

Specific Objectives	Statistical Treatment	Purpose
Patient Registration and Ticketing Queue	Frequency, Percentage, Weighted Mean	Assess the efficiency and completeness of registration and queue management
Vaccination Scheduling and does Monitoring	Weighted Mean, Frequency	Evaluate the accuracy and timeliness of scheduling and follow-up
Vaccine Inventory Monitoring	Weighted Mean, Frequency	Assess stock tracking and expiration monitoring
SMS Reminders and Reporting Modules	Weighted Mean	Measure the effectiveness and accuracy of notifications and reports
System Quality (ISO/IEC 25010)	Weighted Mean	Evaluate the overall quality characteristics of the system
Conduct User Acceptance Testing (UAT)	Frequency, Percentage, Weighted Mean	Determine the acceptability and usability of the system among users

Statistical Treatment of Data

This study employs descriptive statistical methods because its primary objective is to evaluate and describe the performance, usability, reliability, and user acceptance of the developed system rather than to test hypotheses or determine causal relationships. Since the study involves a small, purposively selected population (eight respondents), inferential statistical techniques are not appropriate as they require larger sample sizes and probability-based sampling to ensure generalizability.

This study focuses on system evaluation within a specific institutional context (Mandaluyong City ABTC), meaning that the results are intended to reflect the actual users of the system rather than to generalize findings to a broader population. Predictive or inferential modeling is therefore

unnecessary, as the goal is system improvement and assessment, not prediction or population inference.

System Design and Development

The system was developed using web-based technologies and a relational database management system due to their accessibility, cost-effectiveness, and suitability for multi-user environments such as the ABTC. Specifically, the system utilizes standard web technologies (HTML, CSS, and JavaScript) for the user interface and a server-side scripting language (e.g., PHP) for application logic, enabling dynamic data processing and interaction. A MySQL relational database management system is employed to ensure structured, reliable, and efficient storage of patient records, vaccination schedules, inventory data, and system logs.

These technologies were selected because they are open-source, widely supported, and require minimal licensing cost, making them highly appropriate for government health facilities with limited financial resources. Additionally, web-based systems are platform-independent and can be accessed through standard web browsers, eliminating the need for specialized software installation and reducing deployment complexity.

In terms of scalability, the system is designed using a modular architecture that allows individual components such as patient registration, queue management, scheduling, inventory monitoring, SMS notification, and reporting to be independently updated or expanded without affecting the entire system.

System Architecture

The system adopts a web-based three-tier architecture. The presentation layer consists of a browser- accessible front-end interface used by ABTC staff across different device types. The application layer implements the core business logic for patient registration, ticket queue management, vaccination scheduling, inventory monitoring, SMS notifications, and reporting. The data layer consists of a relational database that stores structured records of patient information, vaccination transactions, inventory items, user accounts, and system logs.

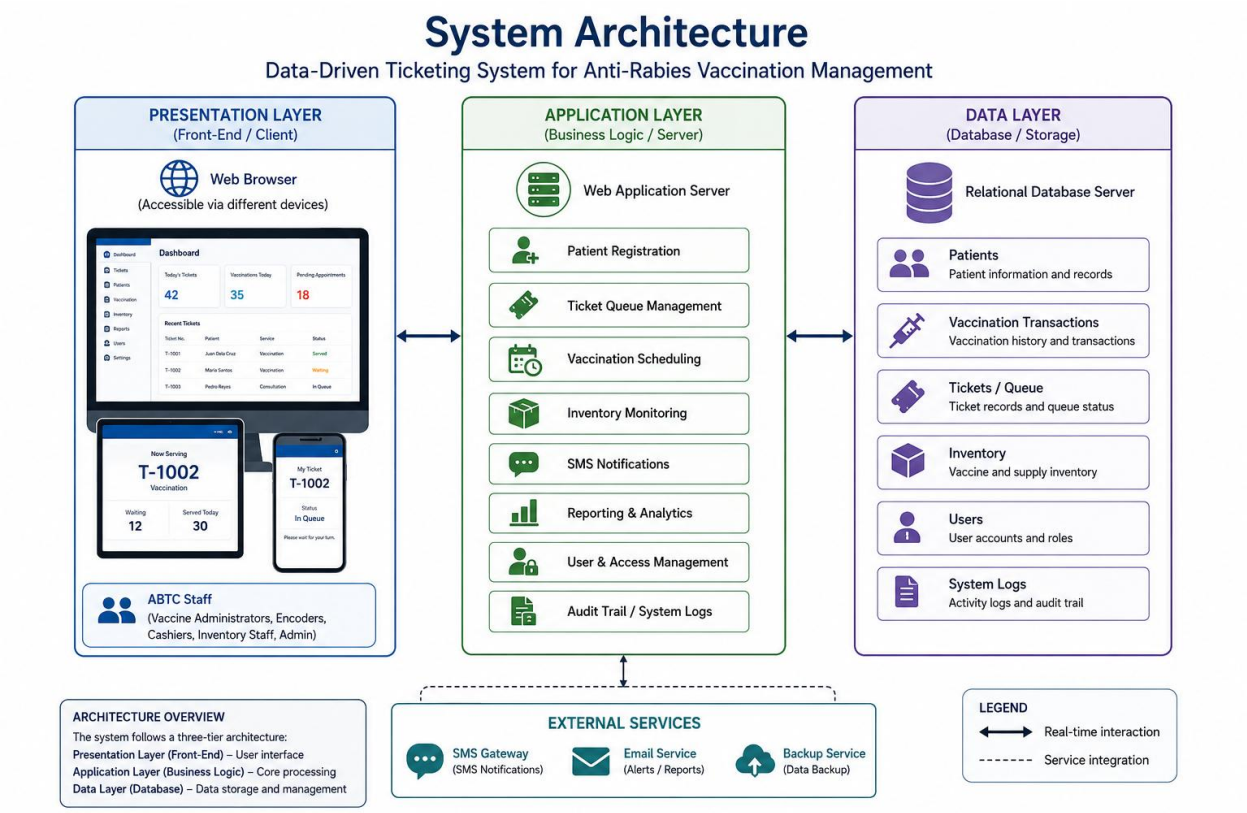


Figure 4.0 System Architecture

Proposed System Overview

Role-based access control is implemented at the application layer to ensure that only authorized personnel can access sensitive patient data, consistent with the requirements of the Data Privacy Act of 2012 (Republic Act No. 10173). The system is developed using open-source frameworks and tools to ensure feasibility within the budget constraints of the Mandaluyong City LGU.

The proposed Data-Driven Ticketing System is composed of six integrated modules designed to streamline ABTC operations: the Patient Registration module, Ticket Queue Management Module, Vaccination Scheduling Module, Vaccine Inventory Monitoring Module, Automated SMS Reminder Module, and Reporting Dashboard.

Each module performs a specific function, such as managing patient information, organizing queue flow, scheduling vaccinations, monitoring vaccine inventory, sending SMS notifications, and

generating reports. These modules work together to ensure a more efficient and organized service delivery within the ABTC.

The system utilized web-based technologies and a relational database management system to ensure efficient data processing, secure storage, and accessibility across different devices. The system is considered data-driven because it continuously collects operational data during ABTC transactions and presents this information through the reporting dashboard to support evidence-based decision making by ABTC administrators and LGU health officers.

Data Flow Diagrams

The Data Flow Diagrams (DFDs) present the logical flow of data within the proposed system. The Context Diagram (Level 0) illustrates the system as a single high-level process and identifies all external entities interacting with it. These include patients, ABTC staff (veterinarians, nurses, and administrative personnel), the SMS gateway, the LGU health office, and the DOH. At the Level 1 DFD, the system is decomposed into its six major processes, each interacting with specific data stores such as the Patient Record, Vaccination Schedule, Inventory Record, Queue Log, Notification Record, and Report Archive.

Selected processes are further decomposed in the Level 2 DFDs to illustrate detailed internal system functionality, including patient registration and profile retrieval, ticket generation and queue management, vaccination scheduling and dose tracking, inventory deduction and alert generation, SMS dispatch, and automated report compilation.

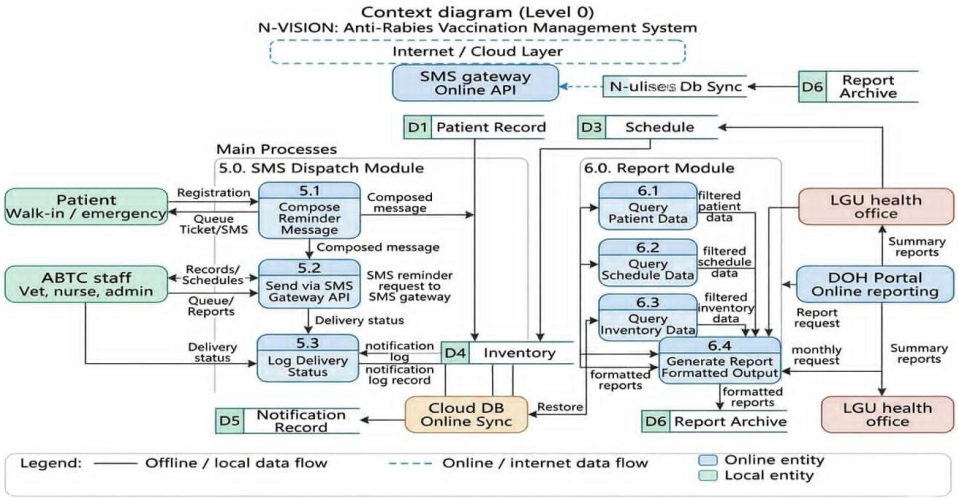


Figure 5.0 Context Diagram (Level 0)

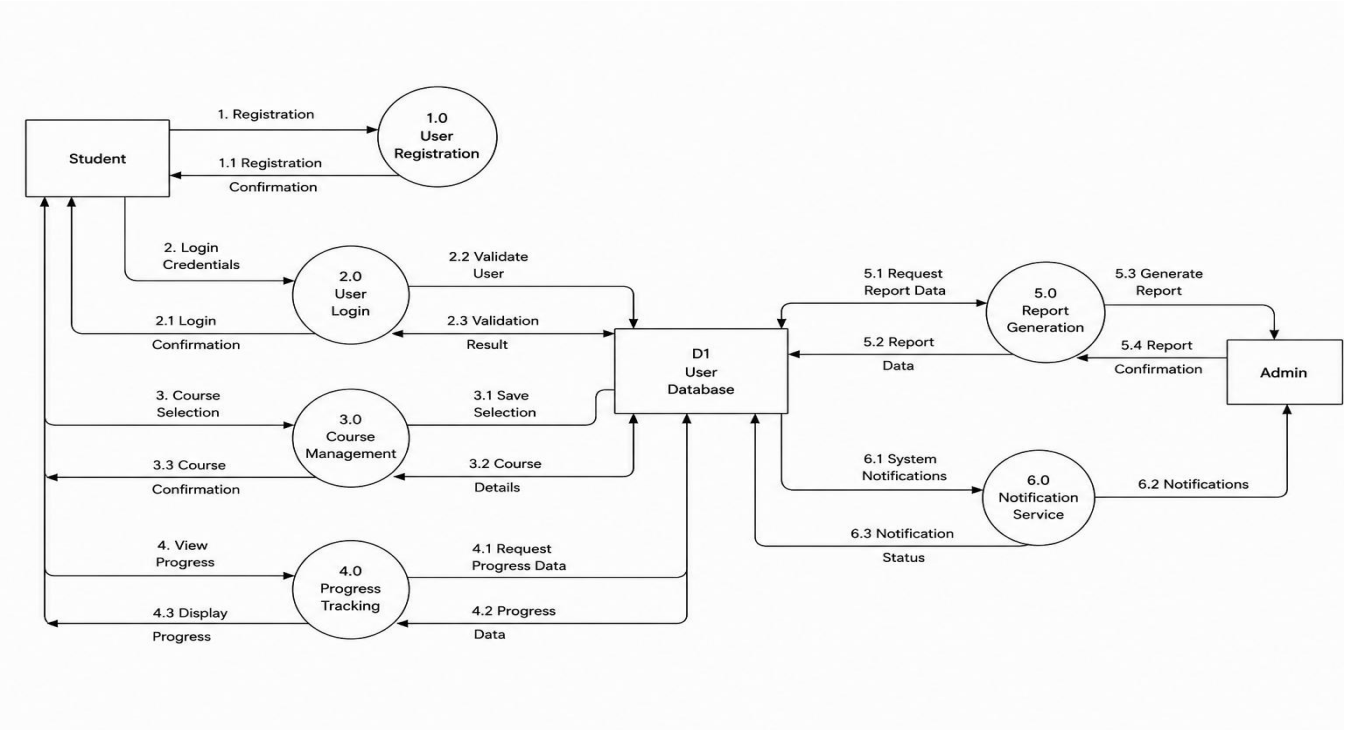


Figure 5.1 Level 1 DFD

Level 2 DFD Decomposition of Selected Process

Patient Registration and Profile Retrieval

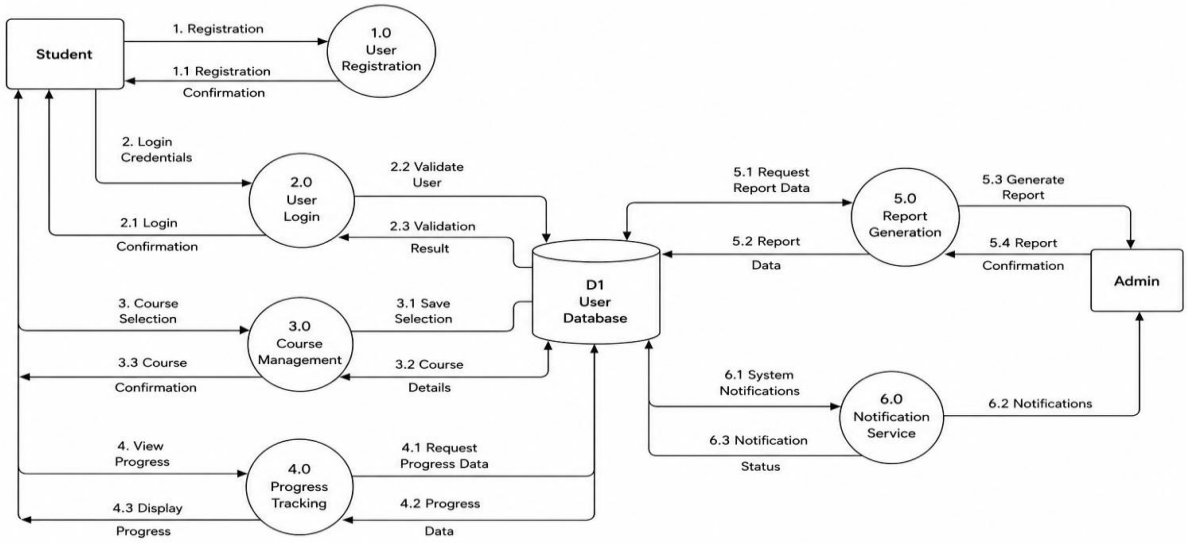


Figure 6.0 Patient Registration and Profile Retrieval

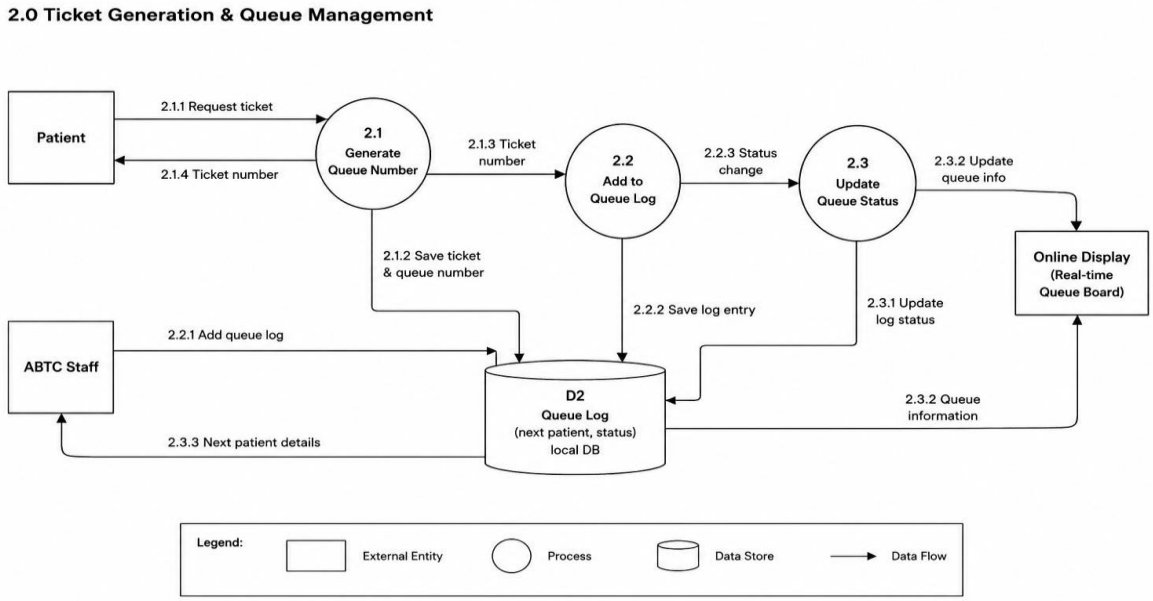


Figure 6.1 Ticket Generation and Queue Management

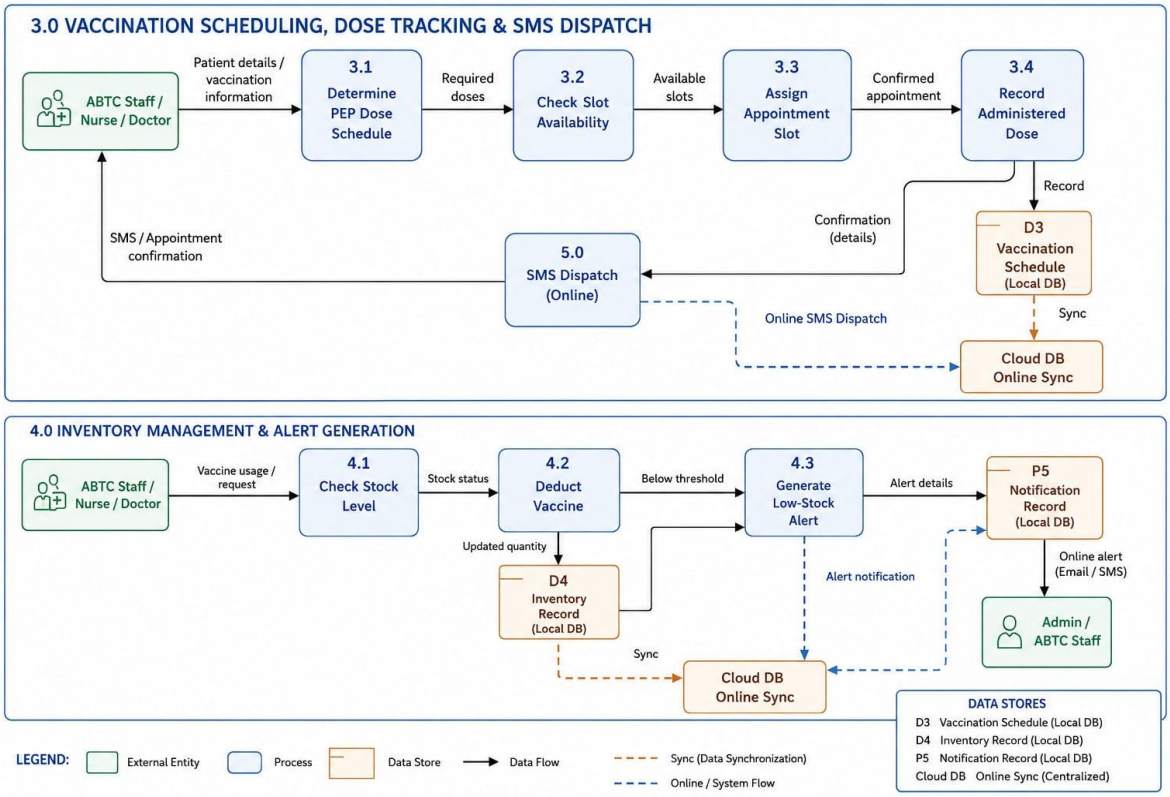


Figure 6.2 Vaccination Scheduling and Inventory Management

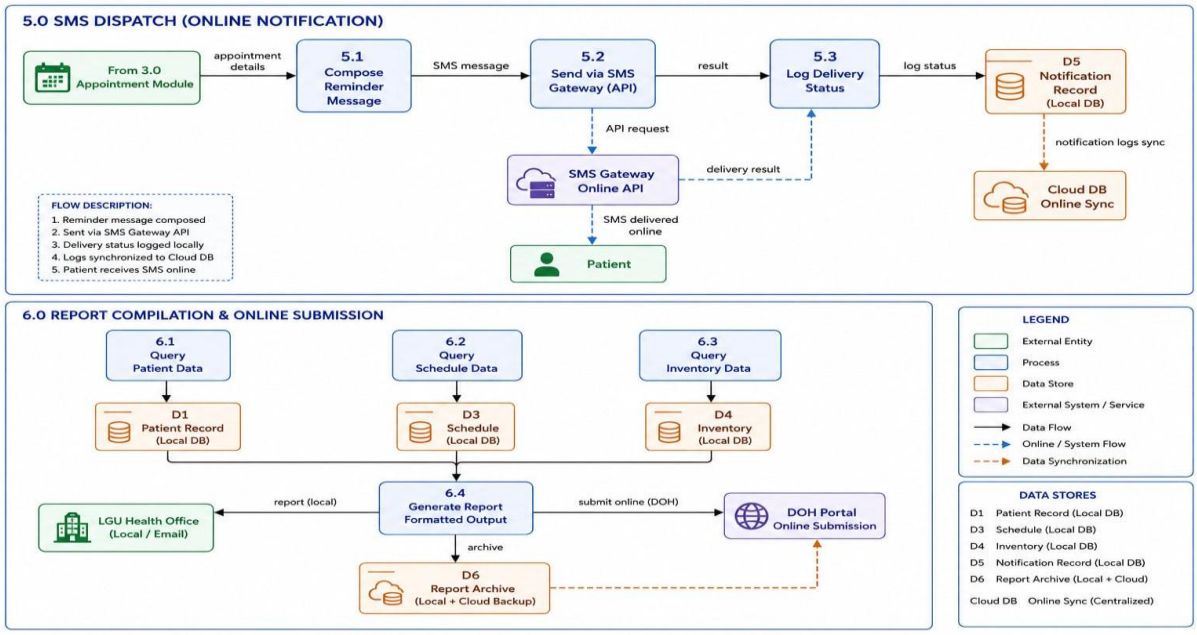


Figure 6.3 SMS Dispatch and Report Compilation10.3 User Interface Design

The user interface is designed to be simple, intuitive, and accessible for ABTC staff with varying levels of digital literacy. Key design principles include the use of clear navigation menus, role-based dashboard views, color-coded queue status indicators, and standardized data entry forms with input validation.

Wireframes for each major module were developed and reviewed by ABTC staff during the requirements analysis phase to ensure alignment with actual clinic workflows prior to full system development.

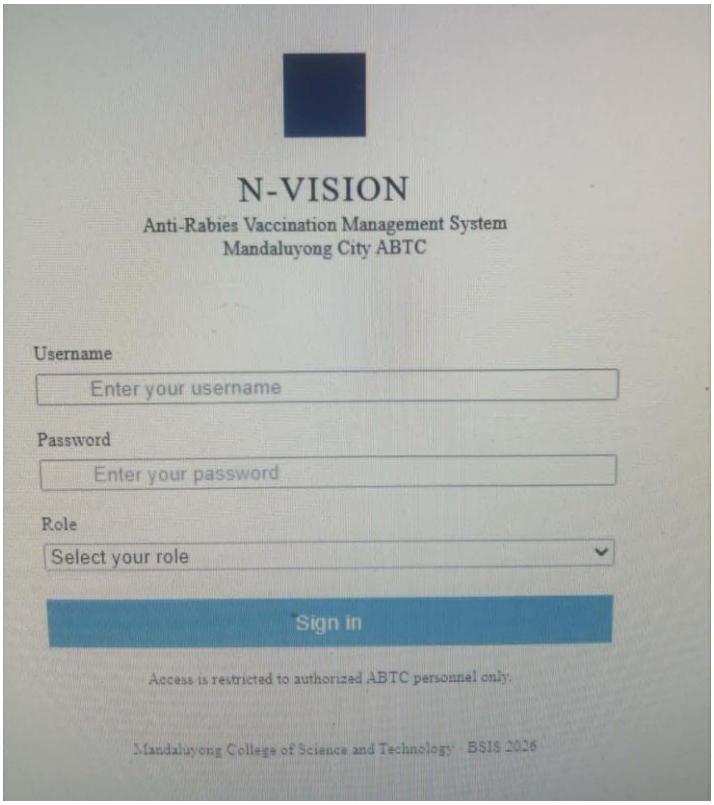


Figure 7.0 Login Screen

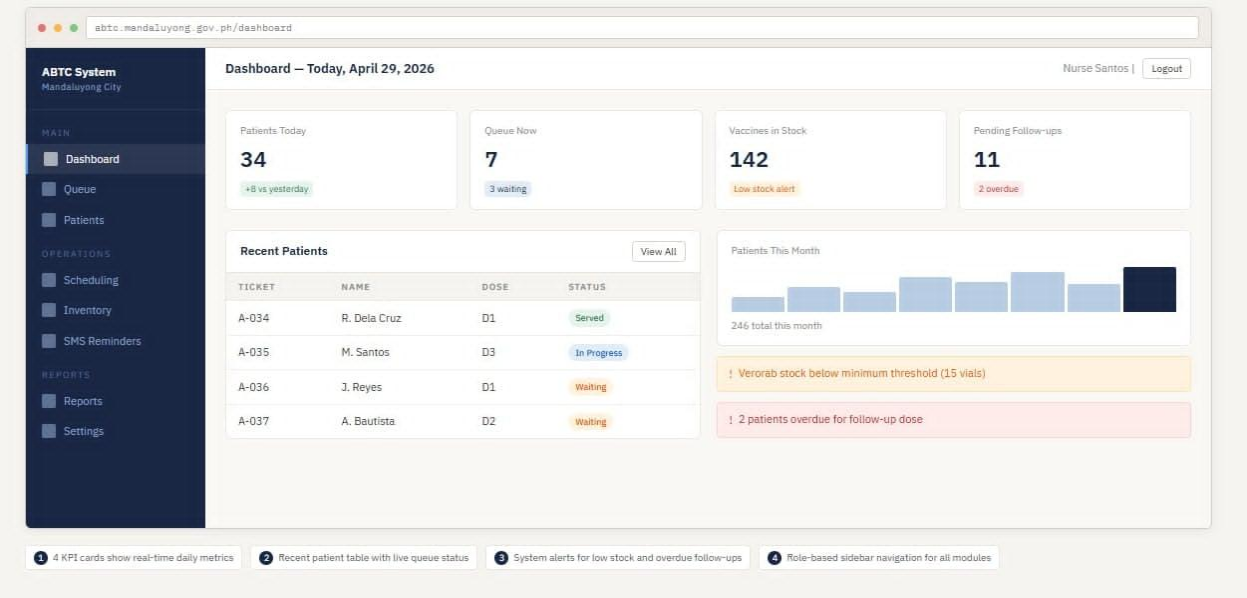


Figure 7.1 Dashboard

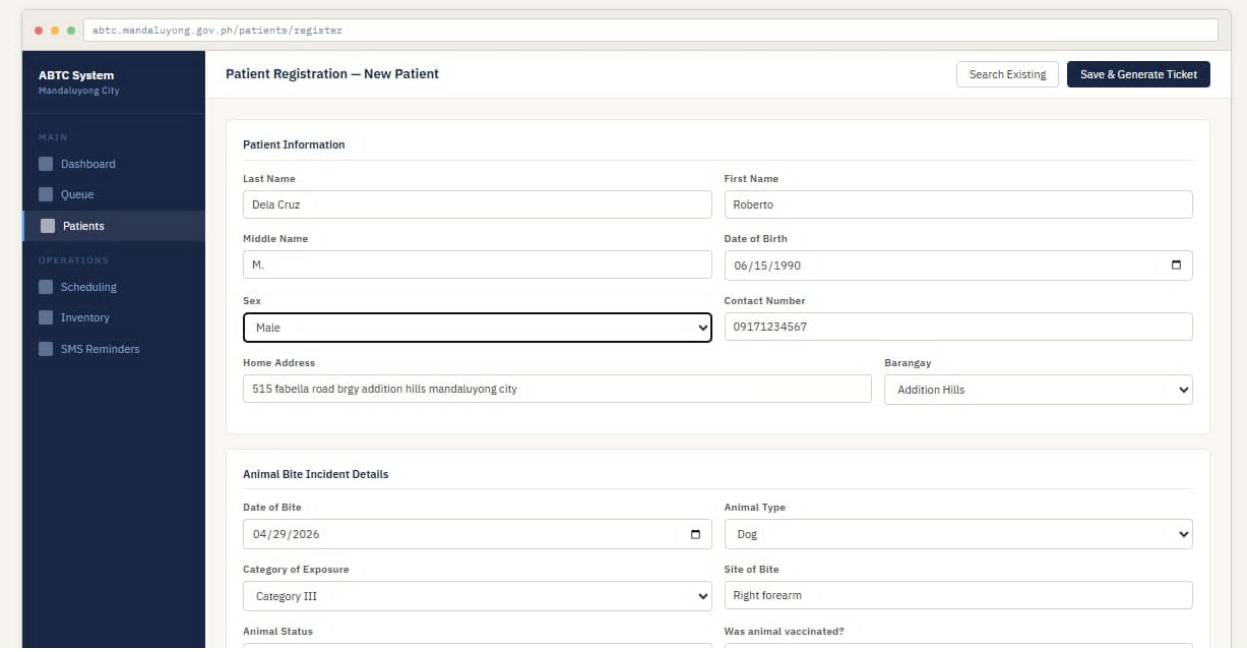


Figure 7.2 Patient Registration Form

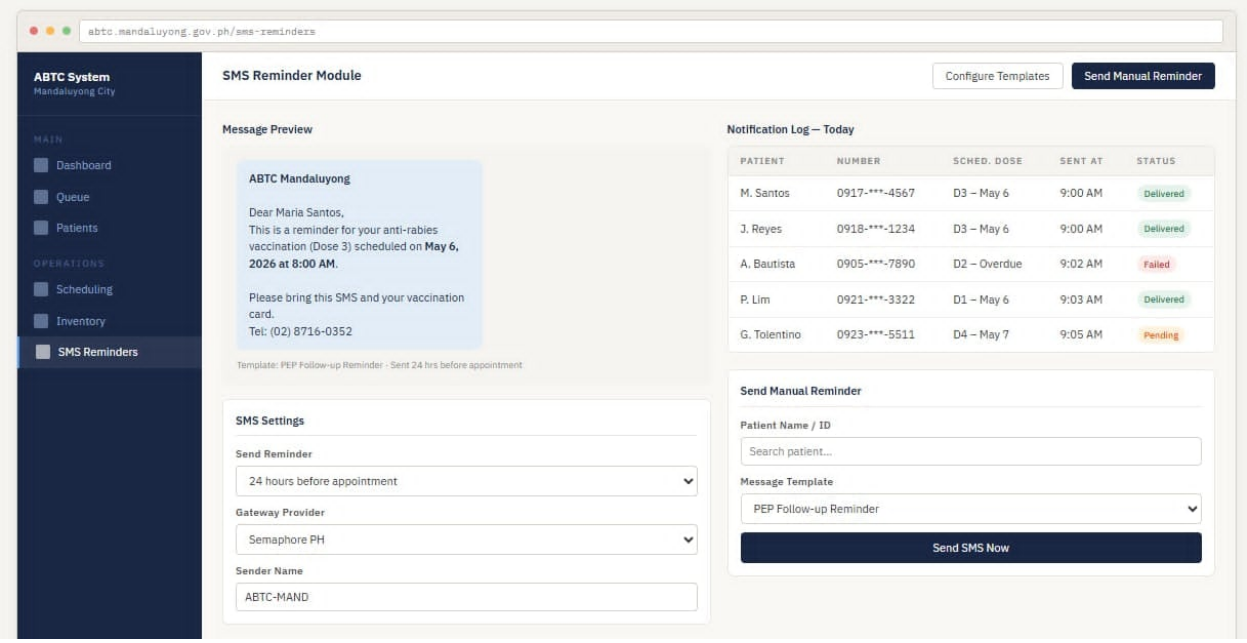


Figure 7.3 SMS Reminder Module

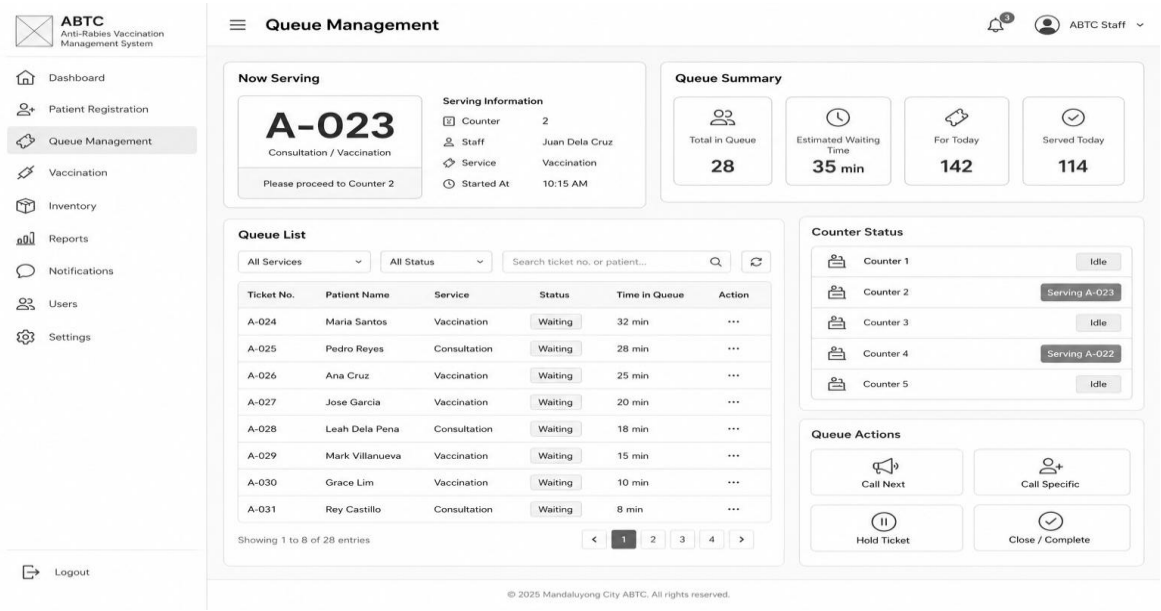


Figure 7.4 Queue Management

Development Process

The development process follows the Agile Scrum methodology described in Section III. Work is organized into six two-week sprints, each delivering an incremental improvement to the system. At the beginning of each sprint, Sprint Planning is conducted to define the sprint goal and identify backlog items to be implemented. The Development Team implements and unit-tests features during the sprint, while Daily Scrum meetings are used to monitor progress and address impediments.

At the end of each sprint, a Sprint Review is conducted to present the working system increment to ABTC staff and gather feedback, which is incorporated into the Sprint Backlog for subsequent sprints. A Sprint Retrospective follows to identify opportunities for improving the development process.

11. Challenges and Solutions

Limited Internet Connectivity

As noted in the introduction, some ABTC facilities experience unreliable internet connectivity, which can disrupt cloud-dependent system functions. To mitigate this, the system is designed to support partial offline functionality for critical operations such as patient registration and ticket generation, with data synchronized to the central database when connectivity is restored.

Resistance to Digital Adoption

ABTC staff are accustomed to paper-based workflows. To reduce resistance, orientation sessions, hands-on demonstrations, and step-by-step user manuals are prepared. The interface is designed to mirror familiar paper-based forms and processes to facilitate a more intuitive transition to the digital system.

Data Privacy Compliance

The system handles sensitive patient data, requiring strict compliance with the Data Privacy Act of 2012. Role-based access control, password policies, audit trails, and data

encryption are implemented to ensure that patient information is accessible only to authorized personnel and that all system activities are logged for accountability.

Budget and Resource Constraints

The project operates within the budget constraints typical of LGU-managed health facilities. All system components are developed using open-source frameworks and tools, with careful selection of stable, well-supported technologies to minimize licensing costs while ensuring long-term maintainability.

12. Integration and Deployment

The Data-Driven Ticketing System will be deployed on an LGU-managed server and accessed via standard web browsers within the ABTC facilities. Deployment involves configuring the web server, application server, and database; creating user accounts; and assigning role-based access permissions in accordance with the Data Privacy Act and institutional policies.

Initial integration will connect the system with the SMS gateway for automated patient notifications. Where feasible, the system will also be integrated with existing LGU health information systems to enable efficient data exchange and improve reporting accuracy. A pilot deployment phase will be conducted during regular ABTC operations, during which the new system will be used alongside existing manual procedures until stability and user confidence are confirmed.

13. Testing and Validation Strategy

The system undergoes several levels of testing to ensure that it meets functional, performance, security, and usability requirements. During development, unit testing and integration testing are performed for each module to verify that individual functions operate correctly and that modules interact seamlessly across the system. System testing evaluates end-to-end workflows such as patient registration, ticket generation, vaccination scheduling, inventory updates, SMS dispatch, and report generation under realistic ABTC operating

conditions.

For formal validation, the deployed system is assessed using two instruments: (1) an ISO/IEC 25010-based evaluation questionnaire administered to a technical expert to assess functional suitability, performance efficiency, usability, reliability, and security, and (2) a UAT questionnaire administered to ABTC staff to evaluate ease of use, satisfaction, and intention to continue using the system. Descriptive statistics including frequency, percentage, and weighted mean are used to summarize test results.

14. Ethical Considerations

Data Privacy and Confidentiality

All identifiable patient information collected and processed by the system is handled in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173). Patient records are accessible only to authorized ABTC personnel, and data security is enforced through user authentication, role-based access control, and audit logging.

Informed Consent for Evaluation

All respondents who participate in surveys and interviews are fully informed of the purpose of the study, the voluntary nature of their participation, and their right to decline or withdraw at any time without consequence. Respondents are assured that their responses will be treated as confidential, used solely for academic and system evaluation purposes, and reported in aggregated form without identifying individual respondents.

Minimization of Risk

No invasive procedures or clinical interventions are introduced by the study. The primary risk relates to potential data security breaches, which are mitigated through technical safeguards including secure authentication, restricted access controls, and regular system backups, as well as through user awareness training on responsible system use.

Responsible Data Handling

Data collected for research purposes, including survey responses and system logs, are stored securely and used exclusively for the purposes of this study. No identifiable patient data are disclosed in the thesis, research presentations, or other outputs.

Beneficence and Respect for Stakeholders

The system is developed with the express goal of benefiting the Mandaluyong City community by improving the efficiency and quality of anti-rabies vaccination services. Feedback from ABTC staff is treated with respect, and recommendations for system improvement are documented for future development cycles.

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Appendices

Appendix A : Self-Devlopment Questionnaire For ABTC (UAT)

Dear Reapondent,

Thank you very much for using the *Data-Driven Ticketing System for Optimized Anti-Rabies Vaccination*

Management at the Mandaluyong City Animal Bite Treatment Center (ABTC). The project aims to design and develop a comprehensive system that streamlines patient registration, manages ticket queues, schedules vaccination doses, monitors vaccine inventory, sends automated SMS reminders, and generates reports for the LGU and DOH.

In this regard, we would like to seek your cooperation by accomplishing this survey form to help us evaluate the quality and assess the level of acceptability of the application to you as our target user. Your feedback will be used to further improve its features to serve you better. Rest assured that the data gathered through this survey will be treated with utmost confidentiality in accordance with the provisions of RA 10173 or the Data Privacy Act of 2012.

Your participation in this evaluation is greatly appreciated. Thank you for your support.

Instructions: Please read each statement carefully and select the number that best reflects your agreement.

Scale	**Description**
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

Problem 1: Patient Registration and Ticket Queue Management

The system should efficiently register patients and manage ticket queues to reduce waiting time and organize patient flow.

No.	Statement	4	**3**	**2**	**1**
1	The system allows easy input of patients' personal and bite incident information.				
2	The system can store and retrieve patient recordsaccurately.				
3	The system allows proper recording of animal bite casedetails.				
4	The system organizes patient profile information in a clear and structured manner.				
5	The system successfully generates a queue ticket after patient registration.				
6	The system displays the queue status in real time.				
7	The system allows staff to call the next patient efficiently.				
8	The ticket queue system reduces patient waiting time.				

Problem 2: Vaccination Scheduling and Dose Monitoring

The system should accurately schedule vaccination doses and monitor follow-up compliance.

No.	**Statement**	**4**	**3**	**2**	**1**

1	The system automatically schedules follow-up vaccination doses based on the PEP protocol.				
2	The system clearly displays upcoming vaccination schedules for each patient.				
3	The system allows staff to update or reschedule vaccination appointments easily.				
4	The system accurately tracks which doses have beenadministered.				
5	The system identifies patients with missed or incomplete vaccination schedules.				
6	The system provides alerts for upcoming and overdue vaccination appointments.				

Problem 3: Vaccine Inventory Monitoring

The system should effectively monitor vaccine stock levels and expiration dates.

No.	**Statement**	**4**	**3**	**2**	**1**
1	The system accurately displays current vaccine stocklevels.				

2	The system deducts vaccine units automatically after each dose is recorded.				
3	The system generates alerts when vaccine stock falls below the minimum threshold.				
4	The system monitors and displays vaccine expirationdates.				
5	The system allows authorized staff to update inventoryrecords easily.				

Problem 4: SMS Reminder and Reporting Module

the system should send automated SMS reminders to patient and generate accurate report 61for the LGU and DOH.

No.	**Statement**	**4**	**3**	**2**	**1**
1	The system sends SMS reminders to patients before their scheduled vaccination appointments.				
2	The SMS reminders contain accurate and complete appointment information.				
3	The system generates vaccination summary reportsaccurately.				
4	The reporting dashboard presents data in a clear and organized manner.				
5	The system allows authorized users to generate and export reports for LGU/DOH submission.				
6	The system reduces the time needed for manual reportpreparation.				

Appendix B: ISO/IEC 25010:2011

Evaluation Questionnaire Dear

Respondent,

Thank you very much for using the *Data-Driven Ticketing System for Optimized Anti-Rabies Vaccination*
Management at the Mandaluyong City Animal Bite Treatment Center (ABTC). The project aims to design and develop a comprehensive system that streamlines patient registration, manages ticket queues, schedules vaccination doses, monitors vaccine inventory, sends automated SMS reminders, and generates reports for the LGU and DOH.

In this regard, we would like to seek your cooperation by accomplishing this survey form to help us evaluate the quality of the system based on ISO/IEC 25010 software quality standards. Your feedback will be used to further improve its features to serve you better. Rest assured that the data gathered through this survey will be treated with utmost confidentiality in accordance with the provisions of RA 10173 or the Data Privacy Act of 2012.

Your participation in this evaluation is greatly appreciated. Thank you for your support.
Instructions:

Please read each statement carefully and select the number that best reflects your agreement.

Scale	Description
4	Strongly Agree (Highly Acceptable)
3	Agree (Acceptable)
2	Disagree (Less Acceptable)
1	Strongly Disagree (Not Acceptable)

PART 1: RESPONDENT'S PROFILE

Name:

Position/Designation:

PART 2: SOFTWARE QUALITY EVALUATION

Functional Suitability - The degree to which the system provides functions that meet stated and implied needs.

No.	Statement	4	3	2	1
1	the system provides all the features need to register patient and manage queues				
2	the system accurately records vaccination schedules and administrated does				
3	the system correctly tracks vaccine inventory levels and expiration date				
4	the system generate correct automated SMS reminders based on schedule appointment				
5	the system produces accurate reports LGU and DOH submission				

Performance Efficiency - The performance relative to the amount of resources used under stated conditions.

No.	**Statement**	**4**	**3**	**2**	**1**
1	The system responds quickly to user inputs during patientregistration.				
2	The ticket queue updates in real time without noticeabledelay.				
3	The system generates reports within an acceptable timeframe.				
4	The system performs consistently even when multiple users are logged in simultaneously.				

Usability - The degree to which the system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction.

No.	Statement		4	3	2
1	The system interface is easy to understand and navigate.				
2	The system is accessible to staff with varying levels ofdigital literacy.				
3	Error messages in the system are clear and helpful.				
4	The system layout and design support efficient taskcompletion.				
5	The system provides role-based dashboard appropriate to each user type				

Reliability - The degree to which the system performs specified functions under specified conditions for a specified period of time.

No.	Statement	4	3	2	1
1	The system operates without unexpected errors or crashes during normal use.				
2	Patient records and vaccination data are stored reliably without data loss.				
3	The system maintains consistent performance across extended operating periods.				
4	The system recovers data correctly after an interruption or connectivity issue				

No.	Statement	4	3	2	1
1	The system restricts access to patient records based on user roles.				
2	The system requires authentication before accessing any module.				
3	The system logs user activities for audit and accountability purposes.				
4	Patient data is protected in accordance with the Data Privacy Act of 2012 (RA 10173).				